

Hossam Abu-elez Fathy

Machine Learning Engineer | Data Analyst

Bachelor of Computer Science & Artificial Intelligence

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🌐 LinkedIn

PROFILE

Computer Science and Artificial Intelligence student with hands-on experience in Machine Learning, Data Analysis, and Software Engineering. Built end-to-end AI projects involving neural networks, optimization algorithms, and full-stack applications. Seeking internships and entry-level opportunities in Machine Learning and Data Science.

EDUCATION

•Artificial Intelligence Sciences

New Mansoura University

Oct 2023 – Jun 2027

CGPA: 3.5 / 4.0

EXPERIENCE

•Machine Learning Trainee

Creativa Innovation Hub

Jun 2025 – Aug 2025

Mansoura

- Preprocessed datasets using Pandas and NumPy.
- Built and evaluated machine learning models using Scikit-Learn.
- Applied classification and regression algorithms.
- Performed feature engineering and model evaluation.
- Visualized insights using Matplotlib and Plotly.
- Collaborated on real-world machine learning projects.

TECHNICAL SKILLS

Programming: Python, C++

Machine Learning: Supervised Learning, Unsupervised Learning, Deep Learning, Neural Networks, Machine Learning Algorithms, Model Evaluation, Model Selection, Feature Engineering, Data Preprocessing

Libraries: Scikit-Learn, Pandas, NumPy, Matplotlib, Plotly

Data Analysis: Excel, Power BI, EDA, Data Cleaning, Data Visualization

Databases: SQL, MySQL, PostgreSQL

Software Engineering: REST APIs, Git, GitHub, Object-Oriented Programming, Software Engineering Methodologies, Full-Stack Development

Tools: Jupyter Notebook, VS Code, Kaggle

Soft Skills: Communication, Team Collaboration, Presentation, Problem Solving

PROJECTS

•MedOptima AI Platform

AI Healthcare Decision Support System

- Developed an end-to-end AI healthcare platform using Neural Networks, Knowledge Based Systems, and Optimization Algorithms.
- Achieved 99.46% test accuracy on a Kaggle dataset containing 41 diseases and 117 symptoms.
- Performed data preprocessing, feature engineering, model evaluation, and hyperparameter tuning.
- Developed REST APIs and an interactive dashboard for diagnosis and analytics.
- Applied software engineering methodologies to build a scalable full-stack application.
- **Technologies:** Python, Scikit-Learn, REST API, SQL, Neural Networks, Full-Stack Development.

•Air Quality Prediction using Machine Learning

Machine Learning Regression Project

2024 – 2025

- Collected and cleaned environmental datasets.
- Performed feature engineering and exploratory data analysis (EDA).
- Built machine learning regression models using Python.
- Evaluated model performance using multiple evaluation metrics.
- Created data visualization dashboards using Plotly and Matplotlib.

•AI Optimization Research Project - FTT Algorithm <i>Research and Optimization</i> – Designed and implemented the Football Team Training (FTT) Optimization Algorithm. – Solved supply chain optimization problems. – Compared algorithm performance with existing optimization methods. – Improved convergence efficiency and solution quality.	2025
•Airline Reservation System <i>Full Software Engineering Project</i> – Designed system requirements, use-case diagrams, class diagrams, and sequence diagrams. – Applied UML methodologies and Object-Oriented Programming principles. – Built the system architecture and enhanced the project using database integration. – Applied software engineering methodologies throughout development.	2023 – 2024
•ALU 4-bit Digital Design Project <i>Computer Architecture Project</i> – Designed and implemented a 4-bit ALU supporting arithmetic and logical operations. – Used digital logic and control signals to perform arithmetic and logical computations. – Simulated and tested system behavior using logic design tools.	2024
•Computer Architecture & Assembly - 8085 & 8086 <i>Assembly Programming</i> – Built a strong understanding of low-level operations and instruction-level execution. – Applied assembly instructions in simulations and step-by-step trace analysis. – Developed a deeper understanding of processor architecture and memory operations.	2024

CERTIFICATES

•Machine Learning Training Creativa Innovation Hub	Jun 2025
•Transact SQL Queries Using SQL Server Mahara Tech	Mar 2025

VOLUNTEERING

•Student Mentor (AI & Programming) Supporting students and beginners	Present
– Helped students and beginners understand AI, programming fundamentals, and problem-solving concepts. – Assisted learners in improving technical skills through mentoring and guidance.	
•Tech Event Organizer AI and Technology Workshops	Present
– Helped organize and coordinate AI and technology workshops for students. – Managed logistics and contributed to creating a smooth learning experience.	

LANGUAGES

- **Arabic:** Native
- **English:** Proficient